# Progress Report 1 Working Plan

Neurosymbolic Verification of LLM Generated Clinical Discharge and Treatment Plans

| Project information | Details |
|---|---|
| Course / team | DATA 298A MSDA Project I \| Team 6 \| Section 22 \| Topic 12 |
| Team members | Akashkumar Senthilkumar, Rajesh Paruchuri, Shruthi Thirukumaran, Sneha Singh, Ritika Mukesh Neema |
| Reporting period | September 10–24, 2026 |
| Date of report | September 24, 2026 |
| Next reporting cycle | September 25–October 8, 2026 |

## Project status

The project is progressing through the requirements, access, and evaluation-design stage. The team has completed required training, clarified the DATA 298A scope, and defined the verifier's principal metrics. Akashkumar Senthilkumar is currently the only member with dataset access; the other four access requests were rejected and have been resubmitted. The team plans to use the dataset Akash downloads as the shared working dataset moving forward. No model-training, data-cleaning, baseline-reproduction, or evaluation result is reported as complete in this period. Percentages below are team estimates based on the current work plan, not audited measurements.

| Workstream | Planning | Investigation | Complete | Documented | Current status |
|---|---|---|---|---|---|
| Project scope and requirements | 90% | 75% | 70% | 65% | Scope and DATA 298A/298B boundary defined |
| Data access and management | 80% | 65% | 55% | 30% | Akash approved; four requests rejected and resubmitted |
| Data engineering | 60% | 30% | 15% | 10% | Dataset identity and file manifest require confirmation |
| Model design and evaluation | 70% | 40% | 25% | 15% | Model A and metric plan drafted; implementation not started |
| Baseline reproduction | 50% | 25% | 0% | 5% | Protocol needs exact task, data, prompt, and scoring alignment |
| Prototype and reproducibility | 35% | 10% | 0% | 0% | Repository exists; pipeline and CI are not yet verified |
| Overall planning estimate | — | — | 30% | 20% | Early-stage progress; evidence collection is the immediate priority |

## Minutes of meetings and coordination

The source materials document a proposed September 22 working meeting and ongoing access coordination, but they do not contain confirmed attendance or signed minutes. To keep this report accurate, the meeting record below is limited to documented coordination items. The team should add actual attendance, decisions, and evidence links before submission if those records are available.

| Date / medium | Engagement and productivity | Issues identified | Recommended follow-up |
|---|---|---|---|
| Sep. 22, 2026 Proposed Zoom session | Agenda covered dataset access, project scope, metrics, model feasibility, and ownership of next actions. | Attendance and final decisions are not confirmed in the supplied record. | Replace this row with verified minutes and attendance; link each decision to the corresponding Linear issue. |
| Sep. 10–24, 2026 Repository / access coordination | All five members completed required training. Akash was approved; the other four requests were rejected and resubmitted. The team agreed to use Akash's downloaded dataset as the shared working dataset. | Four members are awaiting decisions on their resubmitted requests; the shared dataset must be verified from Akash's download. | Track the resubmitted requests, verify Akash's source/version/file manifest and integrity, and use the shared download for team analysis once the checks are complete. |

## Work completed during this period

- All five team members completed the required training for the planned PhysioNet workflow.
- Akashkumar Senthilkumar is the only member currently approved for dataset access and is downloading approximately 9.9 GB of data. The team will use Akash's verified download as the shared working dataset moving

forward, after the source, version, file list, and integrity are confirmed. The materials do not yet establish whether the download includes both MIMIC-IV and MIMIC-IV-Ext-CDM, so the data-engineering milestone remains in progress.

- Ritika Mukesh Neema, Rajesh Paruchuri, Shruthi Thirukumaran, and Sneha Singh received rejected access decisions and resubmitted their requests asking for dataset access. Their individual access remains pending.
- The team clarified the proposed verifier: team-authored clinical rules will be used to check LLM-generated plans, with plan validity as the primary metric and constraint-violation escape rate, repair success, translation fidelity, and solver overhead as secondary measures.
- The abstract established the DATA 298A deliverables: the verifier, rule set, evaluation harness, benchmark, Model A, and reproduction of the Hager et al. baseline. Models B–D are reserved for DATA 298B.
- The project repository and Linear board were identified for code, task ownership, and evidence tracking.

## Issues, risks, and recommended solutions

| Issue or risk | Impact | Owner | Recommended solution and target |
|---|---|---|---|
| Four members received access rejections and have resubmitted their requests. | The team must coordinate through Akash's approved access and shared verified download while the resubmitted requests are reviewed. | Each member; Akash tracks | Track the resubmitted requests; Akash verifies the download and shares the approved working dataset through the team's authorized workflow. |
| The 9.9 GB download is not yet identified by source, version, file list, or checksum. | The team cannot claim that the required datasets are available or reproducible. | Akash | Create a manifest, record source/version and compressed versus extracted size, inventory tables, and preserve raw files. |
| Hager et al. reports diagnostic accuracy, while this project's primary metric is plan validity. | The percentages are different tasks and must not be presented as directly comparable. | Rajesh | Prepare a task/metric mapping and request instructor feedback before finalizing the comparison protocol. |
| Model, compute, and deployment choices remain open. | Model A and milestone feasibility may change. | Shruthi / Ritika | Document candidate model, memory needs, hardware, dependencies, and a restricted-access execution option by Sep. 25. |

## Next reporting period plan

| Owner | Deliverable | Target date | Evidence of completion |
|---|---|---|---|
| Akash | Dataset manifest, schema inventory, integrity checks, and initial ingestion log | Sep. 22–28 | Source URL, version, files, checksums, table/column inventory, sample-load log |
| Ritika | Model A input/output specification and feasible model configuration | Sep. 25 | Written specification and selected configuration |
| Sneha | Initial team-authored rule catalog and valid/invalid synthetic cases | Sep. 27 | Rule IDs, rationale, required inputs, and test examples |
| Rajesh | Metrics, fixed splits, leakage checks, and baseline reproduction protocol | Sep. 29 | Protocol mapping diagnostic accuracy separately from plan validity |
| Shruthi | Pinned environment, reproducible run commands, CI checks, and compute/deployment notes | Sep. 25 | Environment file, commands, and successful synthetic run |
| All members | Replace estimates with verified evidence and complete Workbook 1 sections 1–2 | Oct. 7 | Reviewed workbook, meeting record, artifact links, risks, and schedule |

## Project milestones

| Milestone | Target | Expected result |
|---|---|---|
| Progress Report 1 | Sep. 24, 2026 | Accurate status, work completed, risks, ownership, and next actions |
| Workbook 1 | Oct. 7, 2026 | Requirements, resource plan, literature comparison, risks, and schedule |
| Team Meeting 1 | Oct. 8, 2026 | Demonstrate the data workflow and present metrics, roles, risks, and fixed splits |
| DATA 298A implementation checkpoint | Oct. 22, 2026 | Report actual implementation progress and remaining blockers |

## Project links and references

Repository: https://github.com/Akashkumarsenthil/Clinical-Discharge-TreatmentPlan-Verification

Linear board: https://linear.app/298a-team-6-neurosymbolic/team/298/overview

The project abstract identifies MIMIC-IV and MIMIC-IV-Ext-CDM as the planned data sources, Hager et al. (2024) as the baseline to reproduce, and a target plan-validity rate of at least 0.85 on 100 held-out cases with constraint-violation escape rate at most 0.05. These targets remain future evaluation criteria, not current results.